

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. BPT-502: Classical Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) What is the meaning of constraints? Classify different types of constraints and give at least one example of each.
- (b) The Lagrangian of a two-dimensional system is given by:
- $$L(x, y, \dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - kxy + \frac{1}{2}k_1(x^2 + y^2)$$
- Obtain the Hamiltonian of the system.
- (c) Show that the angular momentum of a particle moving under the influence of a central force is always conserved. What is a cyclic coordinate?
- (d) Construct the Hamiltonian of a system consisting of an electron and a proton. Obtain the canonical equations of motion for both particles.
- (e) Write down the Hamilton-Jacobi equation for a simple harmonic oscillator. Find the solution of this equation.
- (f) Calculate the Poisson bracket $\{\vec{p}^2, \vec{r}\}$.

Question 2.

- (a) Construct the Lagrangian of a simple pendulum of mass m connected to a string of length l . [4]
- (b) How will the Lagrangian change if the string is replaced by a spring of spring constant k and unstretched length l_0 ? [4]
- (c) Obtain the Euler-Lagrange equations of motion for both of the above cases. [6]

Question 3.

- (a) State and explain the virial theorem. Show that the time average kinetic energy $\langle T \rangle$ and time average potential energy $\langle V \rangle$ of a conservative system with a potential $V(\vec{r})$ which is a homogeneous function of coordinates of degree n satisfy:

$$\langle T \rangle = \frac{n}{2} \langle V \rangle$$

[5]

- (b) The Lagrangian for a particle of charge q moving in an electromagnetic field described by scalar potential $\phi(x, y, z, t)$ and vector potential $\vec{A}(x, y, z, t)$ is given by:

$$L = T - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}$$

where $T = \frac{1}{2}m\vec{v}^2$.

- (i) Find the equations of motion for the particle and obtain the Lorentz force law. [5]
 (ii) Calculate the Hamiltonian of this particle. [4]

Question 4.

- (a) A particle of mass m is moving on a plane under an attractive central force given by $F(r) = -\frac{\alpha}{r^2}$ (where $\alpha > 0$). Find the condition for the particle to move in a stable circular orbit. [7]
 (b) A particle moves on a curve $r^2 = a \cos(2\theta)$ under the influence of a central force. Find the force law and show that the areal velocity of the particle is conserved. [7]

OR

- (a) What is the meaning of the isotropy of space? Obtain the conservation law that arises from it. [5]
 (b) Show that if $F(q_i, p_i, t)$ and $G(q_i, p_i, t)$ are two integrals of motion, then their Poisson bracket $\{F, G\}$ is also an integral of motion. [5]
 (c) Discuss the principle of mechanical similarity. [4]

Question 5.

- (a) Determine the values of constants α and β so that the equations:

$$Q = q^\alpha \cos(\beta p), \quad P = q^\alpha \sin(\beta p)$$

represent a canonical transformation. [4]

- (b) Find a suitable generating function for this canonical transformation. [4]
 (c) Show that the fundamental Poisson brackets are invariant under canonical transformations. [6]

OR

- (a) What is virtual work? State and explain D'Alembert's principle. Compare the equations of motion obtained from D'Alembert's principle with those from Hamilton's principle. [7]
 (b) What is meant by a Legendre transformation? Show how it can be used to generate thermodynamic potentials. Derive Hamilton's equations of motion. [7]

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